Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 1.7

1. Determine if the vectors are linearly independent. Justify your answer.

$$\begin{bmatrix} 7 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ 3 \\ -6 \end{bmatrix}, \begin{bmatrix} 8 \\ 6 \\ -24 \end{bmatrix}$$

The vector equation has only the trivial solution, so the vectors are linearly independent.

2. Determine if the vectors are linearly independent.

$$\mathbf{v}_1 = \begin{bmatrix} 5 \\ -2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -5 \\ 2 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** The vectors are not linearly independent because if $c_1 =$ 1 and $c_2 = 1$, both not zero, then $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0}$.
- **B.** The vectors are linearly independent because the vector equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 = \mathbf{0}$ has only the trivial solution.

3. Determine if the columns of the matrix form a linearly independent set. Justify your answer.

$$\begin{bmatrix} 0 & -3 & 9 \\ 2 & 1 & -7 \\ -1 & 4 & -5 \\ 1 & -4 & -2 \end{bmatrix}$$

Select the correct choice below and fill in the answer box within your choice.
(Type an integer or simplified fraction for each matrix element.)

- **A.** If A is the given matrix, then the augmented matrix _____ represents the equation $A\mathbf{x} = \mathbf{0}$. The reduced echelon form of this matrix indicates that $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Therefore, the columns of A do not form a linearly independent set.
- **B.** If A is the given matrix, then the augmented matrix _____ represents the equation $A\mathbf{x} = \mathbf{0}$. The reduced echelon form of this matrix indicates that $A\mathbf{x} = \mathbf{0}$ has more than one solution. Therefore, the columns of A form a linearly independent set.
- **C.** If A is the given matrix, then the augmented matrix _____ represents the equation $A\mathbf{x} = \mathbf{0}$. The reduced echelon form of this matrix indicates that $A\mathbf{x} = \mathbf{0}$ has more than one solution. Therefore, the columns of A do not form a linearly independent set.
- **D.** If A is the given matrix, then the augmented matrix $\begin{bmatrix} 0 & -3 & 9 & 0 \\ 2 & 1 & -7 & 0 \\ -1 & 4 & -5 & 0 \\ 1 & -4 & -2 & 0 \end{bmatrix}$ represents the equation $A\mathbf{x} = \mathbf{0}$. The reduced echelon form of this matrix indicates that $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Therefore, the columns of A form a linearly independent set.

4. Determine if the columns of the matrix form a linearly independent set. Justify your answer.

$$\begin{bmatrix} 1 & -3 & 2 & 2 \\ -3 & 9 & -6 & 2 \\ 0 & 1 & -1 & 3 \end{bmatrix}$$

Choose the correct answer below.

- **A.** The columns of the matrix do not form a linearly independent set because there are more entries in each vector than there
- **B.** The columns of the matrix do not form a linearly independent set because the set contains more vectors than there are entries in each vector.
- **C.** The columns of the matrix do form a linearly independent set because there are more entries in each vector than there are vectors in the set.
- **D.** The columns of the matrix do form a linearly independent set because the set contains more vectors than there are entries in each vector.

5. Determine by inspection whether the vectors are linearly independent. Justify your answer.

$$\begin{bmatrix} 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 9 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 6 \end{bmatrix}$$

Choose the correct answer below.

- **A.** The set is linearly dependent because at least one of the vectors is a multiple of another vector.
- **B.** The set is linearly independent because at least one of the vectors is a multiple of another vector.
- **C.** The set is linearly independent because there are four vectors in the set but only two entries in each vector.
- **D.** The set is linearly dependent because there are four vectors but only two entries in each vector.

6. Determine by inspection whether the vectors are linearly independent. Justify your answer.

$$\begin{bmatrix} 8 \\ -6 \\ 4 \end{bmatrix}, \begin{bmatrix} 12 \\ -9 \\ 6 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** The set of vectors is linearly dependent because $\frac{3}{2}$ times the first vector is equal to the second vector.

  (Type an integer or a simplified fraction.)
- **B.** The set of vectors is linearly dependent because neither vector is the zero vector.
- **C.** The set of vectors is linearly independent because neither vector is a multiple of the other vector.

7. Determine by inspection whether the vectors are linearly independent. Justify your answer.

$$\begin{bmatrix} 4 \\ 6 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 3 \\ 3 \end{bmatrix}$$

Choose the correct answer below.

- **A.** The set of vectors is linearly dependent because _____ times the first vector is equal to the third vector. (Type an integer or a simplified fraction.)
- **B.** The set of vectors is linearly independent because _____ times the first vector is equal to the second vector.
- **C.** The set of vectors is linearly dependent because one of the vectors is the zero vector.
- **D.** The set of vectors is linearly independent because none of the vectors are multiples of the other vectors.

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8. Determine by inspection whether the vectors are linearly independent. Justify your answer.

$$\begin{bmatrix} 8 \\ 12 \\ -4 \end{bmatrix}, \begin{bmatrix} -2 \\ -3 \\ -1 \end{bmatrix}$$

Choose the correct answer below.

- **A.** The set is linearly dependent because neither vector is a multiple of the other vector. Two of the entries in the first vector are -4 times the corresponding entry in the second vector. But this multiple does not work for the third entries.
- **B.** The set is linearly independent because neither vector is a multiple of the other vector. Two of the entries in the first vector are -4 times the corresponding entry in the second vector. But this multiple does not work for the third entries.
- **C.** The set is linearly independent because the first vector is a multiple of the other vector. The entries in the first vector are -4 times the corresponding entry in the second vector.
- **D.** The set is linearly dependent because the first vector is a multiple of the other vector. The entries in the first vector are -4 times the corresponding entry in the second vector.

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9. Determine whether the statement below is true or false. Justify the answer.

Two vectors are linearly dependent if and only if they lie on a line through the origin.

Choose the correct answer below.

- **A.** The statement is false. If two vectors are linearly dependent then the graph of one will be orthogonal, or perpendicular, to the other.
- **B.** The statement is true. Two vectors are linearly dependent if one of the vectors is a multiple of the other. Two such vectors will lie on the same line through the origin.
- **C.** The statement is true. Linearly dependent vectors must always intersect at the origin.
- **D.** The statement is false. Two vectors are linearly dependent if one of the vectors is a multiple of the other. The larger vector will be further from the origin than the smaller vector.

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10. Determine whether the statement below is true or false. Justify the answer.

If S is a linearly dependent set, then each vector is a linear combination of the other vectors in S.

Choose the correct answer below.

- **A.** The statement is true. If S is linearly dependent then for each j, $\mathbf{v}_j$, a vector in S, is a linear combination of the preceding vectors in S.
- **B.** The statement is true. If an indexed set of vectors, S, is linearly dependent, then at least one of the vectors can be written as a
- **C.** The statement is false. If an indexed set of vectors, S, is linearly dependent, then it is only necessary that one of the vectors is a linear combination of the other vectors in the set.
- **D.** The statement is false. If S is linearly dependent then there is at least one vector that is not a linear combination of the other vectors, but the others may be linear combinations of each other.